

Calc 3 – Green, Stokes, Divergence

Worksheet A

1. Use Green's theorem to compute $\oint_C (x^2 dy - y^2 dx)$ where C is the unit circle (counter-clockwise).
2. Use Green's theorem to compute the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ using $A = \frac{1}{2} \oint x dy - y dx$.
3. Use Stokes' theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle -y, x, z \rangle$ where C is the boundary of the disk $x^2 + y^2 \leq 1$, $z = 0$, oriented counter-clockwise.
4. Use the divergence theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ through the unit cube $[0, 1]^3$.
5. Use the divergence theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ through the unit sphere.

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Worksheet B

1. Use Green's theorem to compute $\oint_C y^3 dx - x^3 dy$ where C is the unit circle (CCW).
2. Use Green's theorem to evaluate $\oint_C (e^x + y^2) dx + (e^y + x^2) dy$ on the boundary of the rectangle $[0, 1] \times [0, 2]$ (CCW).
3. Use Stokes' theorem to compute $\iint_S (\text{curl } \mathbf{F}) \cdot d\mathbf{S}$ where $\mathbf{F} = \langle yz, xz, xy \rangle$ and S is the upper half-sphere $x^2 + y^2 + z^2 = 1, z \geq 0$, oriented upward.
4. Use the divergence theorem to compute the outward flux of $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$ through the surface of the box $[0, 1] \times [0, 2] \times [0, 3]$.
5. Verify the divergence theorem for $\mathbf{F} = \langle x, 0, 0 \rangle$ on the unit ball (compute both sides).

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Answer Key (Worksheets A & B)

Worksheet A.

1. Green: $\oint(-y^2 dx + x^2 dy) = \iint_D(Q_x - P_y) dA = \iint_D(2x - (-2y)) dA = \iint_D 2(x + y) dA = 0$ (by symmetry). $\boxed{0}$.
2. $A = \frac{1}{2} \int_0^{2\pi} (a \cos t \cdot b \cos t - b \sin t \cdot (-a \sin t)) dt = \frac{ab}{2} \int_0^{2\pi} 1 dt = \boxed{\pi ab}$.
3. $\text{curl } \mathbf{F} = \langle 0, 0, 2 \rangle$. Surface S is the disk in the xy -plane, normal $\mathbf{k}$. Integral $= \iint_S 2 dA = 2 \cdot \pi = \boxed{2\pi}$.
4. $\text{div } \mathbf{F} = 3$. Flux $= 3 \cdot \text{Vol}([0, 1]^3) = \boxed{3}$.
5. $\text{div } \mathbf{F} = 3(x^2 + y^2 + z^2) = 3\rho^2$. Flux $= \int_0^{2\pi} \int_0^\pi \int_0^1 3\rho^2 \cdot \rho^2 \sin \varphi d\rho d\varphi d\theta = 3 \cdot 2\pi \cdot 2 \cdot \frac{1}{5} = \boxed{\frac{12\pi}{5}}$.

Worksheet B.

1. $P = y^3$, $Q = -x^3$. $Q_x - P_y = -3x^2 - 3y^2$. $\oint = -3 \iint_D (x^2 + y^2) dA = -3 \int_0^{2\pi} \int_0^1 r^2 \cdot r dr d\theta = -3 \cdot 2\pi \cdot \frac{1}{4} = \boxed{-\frac{3\pi}{2}}$.
2. $Q_x - P_y = 2x - 2y$. $\oint = \int_0^1 \int_0^2 (2x - 2y) dy dx = \int_0^1 (4x - 4) dx = 2 - 4 = \boxed{-2}$.
3. $\mathbf{F} = \nabla(xyz)$ is conservative, so $\text{curl } \mathbf{F} = \mathbf{0}$. Integral $= \boxed{0}$.
4. $\text{div } \mathbf{F} = 2(x + y + z)$. Flux $= 2 \int_0^1 \int_0^2 \int_0^3 (x + y + z) dz dy dx$. Inner: $3x + 3y + 9/2$. Next: $\int_0^2 (3x + 3y + 9/2) dy = 6x + 6 + 9 = 6x + 15$. Outer: $\int_0^1 (6x + 15) dx = 3 + 15 = 18$. Flux $= 2 \cdot 18 = \boxed{36}$.
5. LHS (volume): $\text{div } \mathbf{F} = 1$, $\iiint_B 1 dV = \frac{4\pi}{3}$. RHS (surface): use spherical, $\mathbf{F} \cdot \mathbf{n} = x \cdot x = \sin^2 \varphi \cos^2 \theta$ on $\rho = 1$. $\iint_S x^2 dS = \int_0^{2\pi} \int_0^\pi \cos^2 \theta d\theta \int_0^\pi \sin^3 \varphi d\varphi = \pi \cdot \frac{4}{3} = \frac{4\pi}{3}$. **Both** $= 4\pi/3$. $\checkmark$